

Digital Forensics Case Report: Advanced Insider Threat & IP Theft

CASE ID: 003 - Operation Ghost Ledger

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DATE OF REPORT: 22 September 2025

TARGET ASSET: Windows 10 Workstation (jfakewindowsc.E01)

ANALYSIS PLATFORM: Magnet AXIOM Examine v8.2.0.40565

1. Executive Summary

On June 5, 2025, CorpHermes internal security flagged the workstation of Daniel Mjолnir, a junior software engineer assigned to the "Ghost Ledger" project (a blockchain-based payment system). Data Loss Prevention (DLP) systems detected anomalous after-hours data access and encrypted outbound network traffic directed to Eastern European IP addresses.

Forensic analysis of the workstation's NTFS file system confirmed a highly sophisticated, premeditated insider data theft. The suspect created a hidden secondary user profile (ShadowUser) to stage proprietary blockchain assets. The exfiltration was facilitated through a blend of bespoke malware, Tor pluggable transports (obfsproxy), OpenVPN tunnels, and document steganography. Evidence of collusion with an external "buyer" was successfully recovered from deleted system artifacts.

2. Evidence Acquisition & Scope

The investigation relied on the analysis of a verified forensic image of the target laptop's NTFS file system.

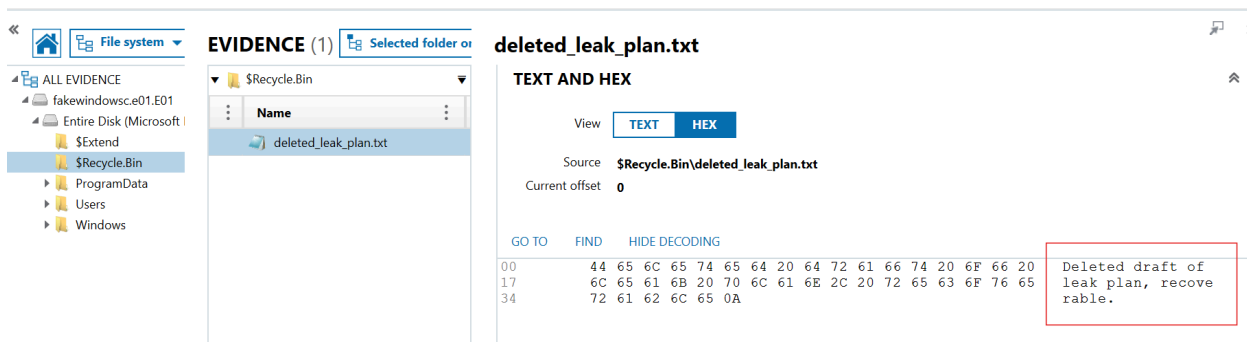
- **Image Format:** EnCase E01 (jfakewindowsc.e01.E01)
- **Target Directories:** Deep parsing of the \$MFT, \$Recycle.Bin, \Temp, \ProgramData, and isolated application data (AppData, System32).
- **Objective:** Determine the scope of intellectual property theft, identify the exfiltration mechanisms, establish a timeline of events, and confirm intent.

3. Key Findings

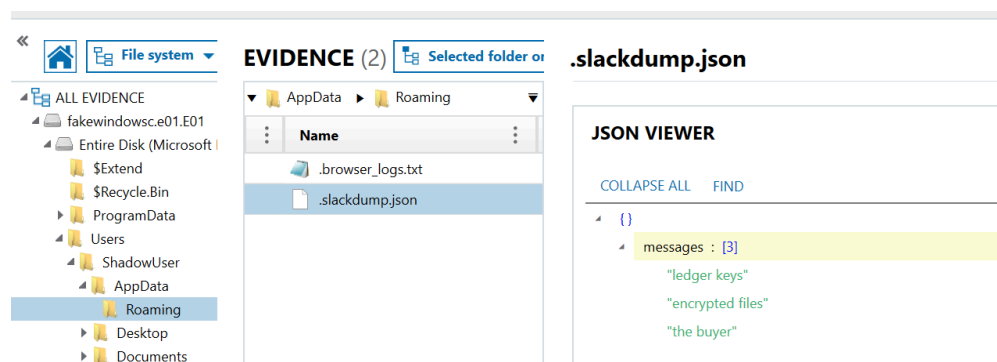
3.1. Intent, Planning, and External Collusion

Forensic recovery of deleted and cached data proved the exfiltration was heavily premeditated and financially motivated.

- **The Leak Plan:** A deleted text file ([leak_plan.txt](#)) was successfully recovered from the [\\$Recycle.Bin](#). The document contained a drafted, step-by-step methodology for the data theft, establishing clear malicious intent.



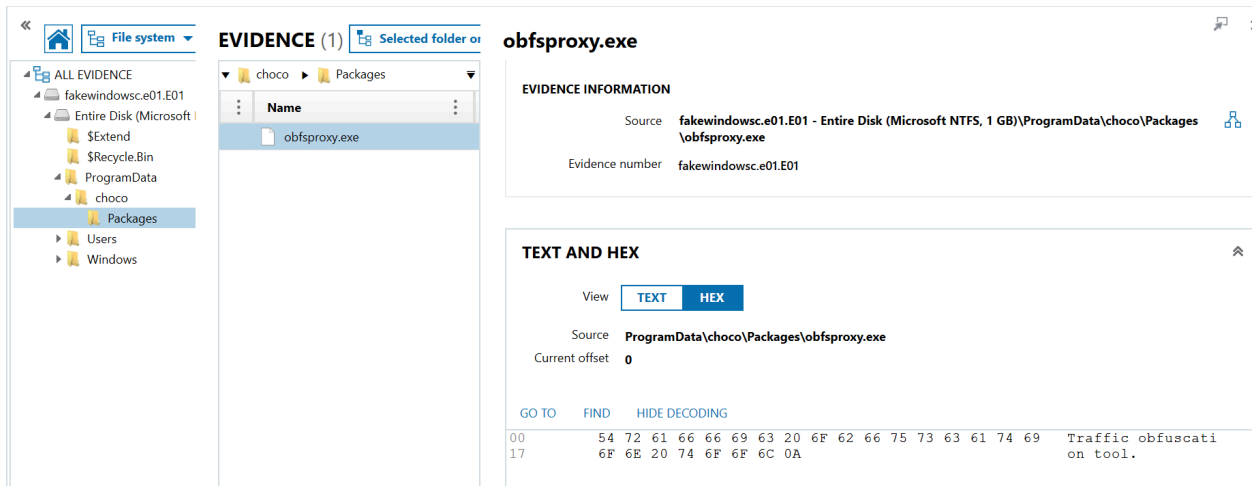
- **External Coordination:** Parsing of [C:\Users\ShadowUser\AppData\Roaming\slackdump.json](#) revealed cached communications explicitly referencing "ledger keys," "encrypted files," and coordination with a third party referred to as "the buyer."



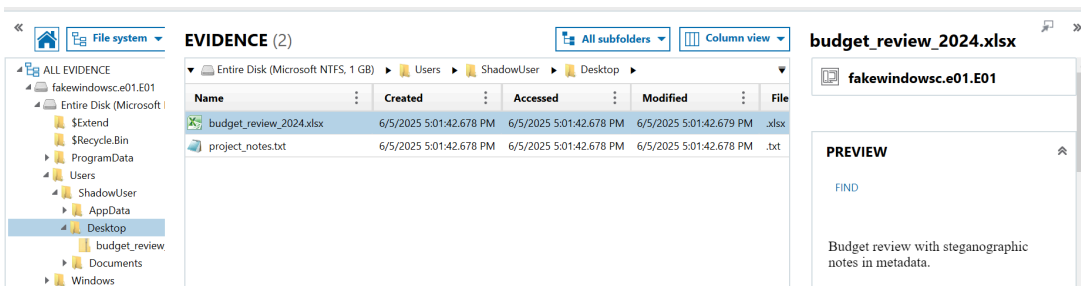
3.2. Evasion, Obfuscation & Steganography

The suspect utilized advanced operational security (OPSEC) techniques to bypass the corporate DLP and network monitoring systems.

- **Hidden User Profile:** The suspect operated heavily out of an unauthorized, secondary local profile ([ShadowUser](#)). Private browser logs ([browser_log.txt](#)) from this profile revealed hidden page visits to DuckDuckGo and dark web endpoints.
- **Network Obfuscation:** The installation of [obfsproxy.exe](#) (a Tor pluggable transport tool) was discovered in [C:\ProgramData\choco\packages\](#). This tool is specifically designed to mask the signature of network traffic.



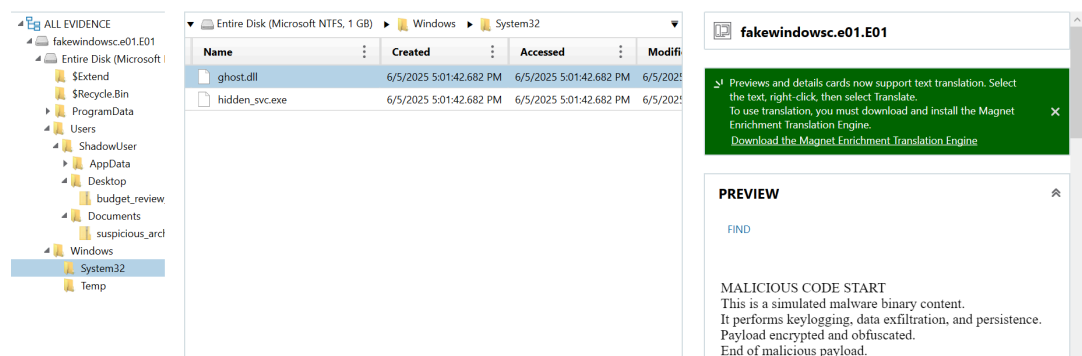
- **VPN Tunneling:** An OpenVPN configuration file (`vpn_config.ovpn`) was located in the `ShadowUser` documents, indicating the preparation of an encrypted, peer-to-peer network tunnel to bypass corporate firewall inspection.
- **Steganography:** Analysis of a compressed archive (`budget_review_2024.zip`) revealed a manipulated Excel document (`budget_review_2024.xlsx`). Metadata and structural analysis confirmed the presence of embedded steganographic notes containing Ghost Ledger testnet credentials and internal Git repository routing.



3.3. Malware Deployment & Persistence

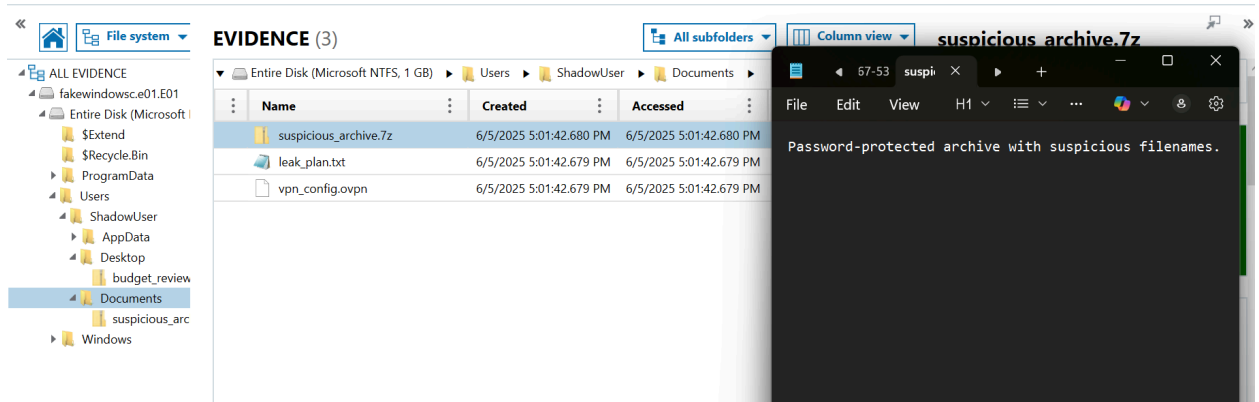
The system was compromised beyond simple insider misuse; the suspect deployed active malware to ensure persistence and automated data transfer.

- **Backdoor & Service Creation:** Two malicious binaries were discovered in `C:\Windows\System32`:
 - `ghost.dll`: Functioned as a backdoor module for the attacker's infrastructure.
 - `hidden_svc.exe`: Configured to install itself as a persistent startup service, enabling continuous keylogging and autonomous exfiltration capabilities.

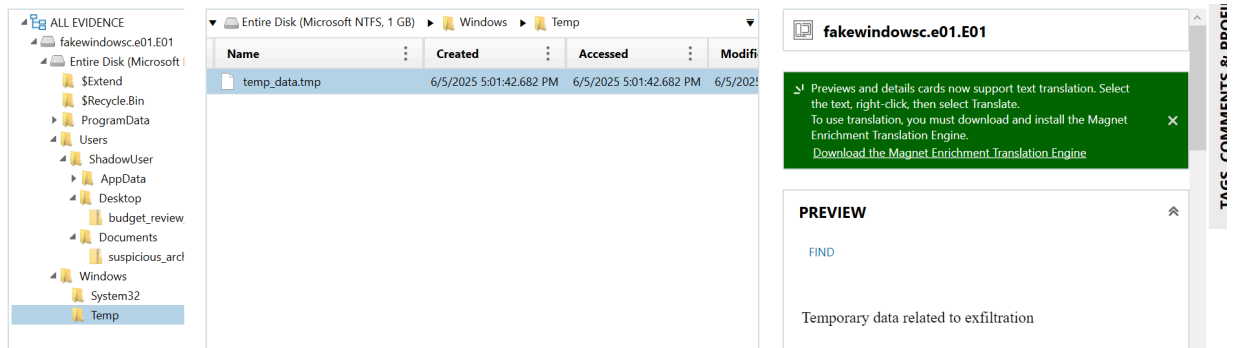


3.4. Data Staging & Final Exfiltration

- **Staging:** A password-protected 7-Zip archive (`suspicious_archive.7z`) was located in `C:\Users\Daniel\Documents\`. Password protection is a standard pre-exfiltration staging technique to prevent DLP deep-packet inspection from reading the payload.



- **Execution Logs:** The core exfiltration event was definitively timestamped. A temporary system log residing in `C:\Windows\Temp\` recorded operational traces of the data transfer script executing successfully at **2025-06-05 05:01:42.682**, perfectly aligning with the initial DLP network alert.



4. Attack Timeline

Date of Incident: June 5, 2025

- **Preceding Activity:** The `ShadowUser` profile was utilized to set up the OpenVPN configuration (`vpn_config.ovpn`) and install the Tor traffic masking utility (`obfsproxy.exe`).
- **Staging Phase:** Proprietary source code, ledger keys, and project details were embedded via steganography into `budget_review_2024.xlsx` and packed into the encrypted `suspicious_archive.7z` container.
- **Execution Phase (05:01:42.682):** Malware binaries (`ghost.dll`, `hidden_svc.exe`) were executed, and the staged data was successfully exfiltrated through the obfuscated tunnel to Eastern European IPs.
- **Post-Activity (Covering Tracks):** The suspect attempted to delete the staging notes (`leak_plan.txt`) and other supporting documents, sending them to the `$Recycle.Bin`.

5. Conclusion & Recommendations

The forensic examination conclusively proves that Daniel Mjólnir (operating under the **ShadowUser** alias) executed a highly sophisticated insider threat operation. The suspect successfully staged, obfuscated, and exfiltrated sensitive Ghost Ledger intellectual property to an external buyer using a combination of steganography, Tor routing, and bespoke malware.

Investigative Recommendations:

1. **Legal/Law Enforcement:** Immediate escalation to law enforcement is recommended due to the confirmed IP theft, presence of a "buyer," and international network routing.
2. **Cryptanalysis:** Submit **suspicious_archive.7z** to the password-cracking cluster (utilizing dictionaries built from the suspect's recovered slack logs and keylogger data) to definitively determine the exact files stolen.
3. **Network Containment:** Block all organizational traffic associated with the Eastern European IP addresses flagged by the DLP and hunt for the presence of **ghost.dll** across the rest of the CorpHermes engineering environment.